

EN401 — Handout 3 Results & Analysis Report

Base Power System Implementation & Backstop Penalty Analysis

1. Scenario Objective & LP Solution Metrics

- **Objective Value (NPV):** \$51,009,521.16
- **Problem Scale:** 466 constraints, 240 variables, 2,160 non-zeros
- **Technologies Active:** MINBACK, BACKSTOP

2. Electricity Demand Trajectory (20 to 90 PJ)

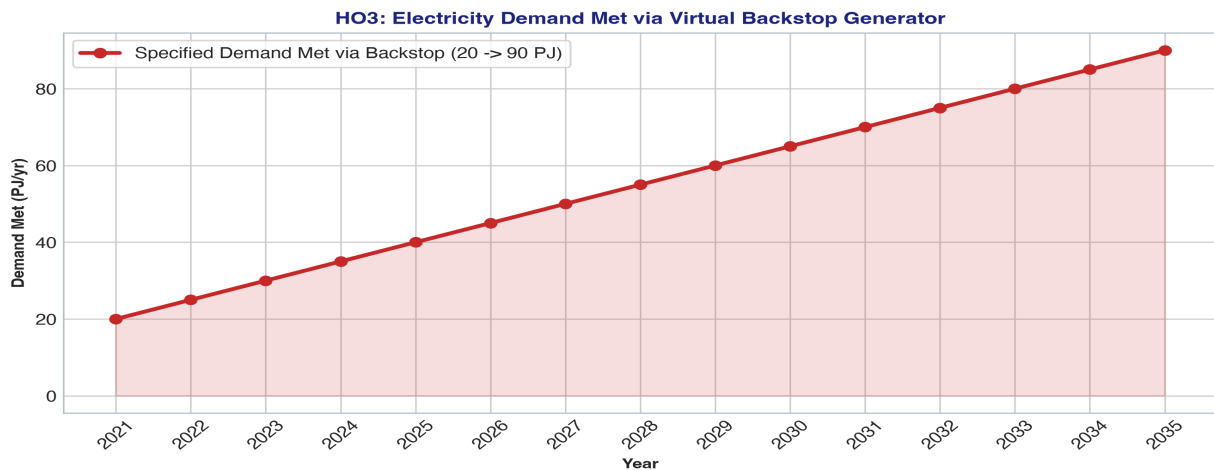


Figure: Final electricity demand met via the virtual backstop generator.

3. Virtual Backstop Capacity Additions

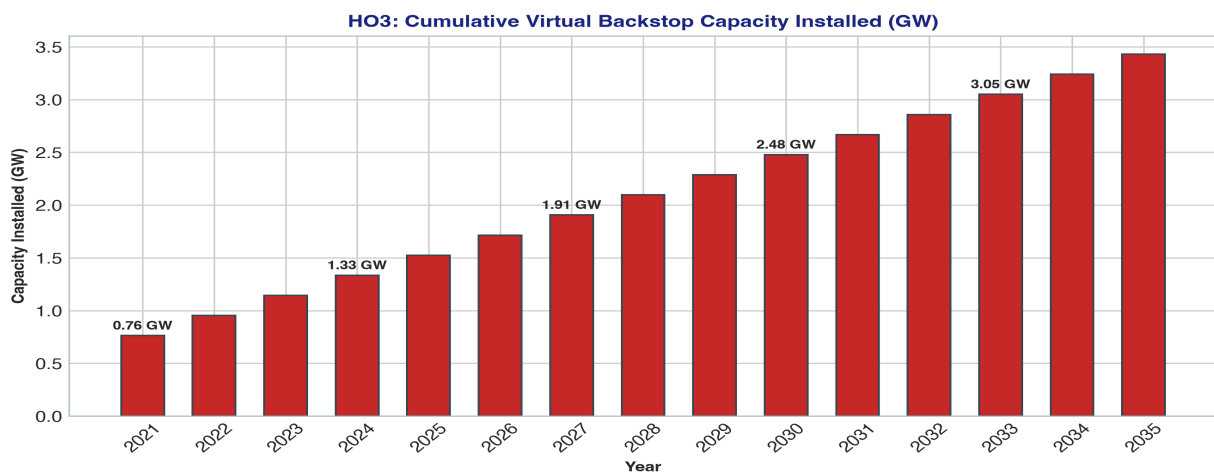


Figure: Cumulative installed capacity of BACKSTOP reaching 3.43 GW by 2035.

4. Economic Logic

The extreme \$51M system cost is driven by the \$99,999/kW capital cost assigned to BACKSTOP. This acts as a mathematical penalty ensuring the model will rapidly switch to realistic power plants once defined.